



Clots from vax antibodies

The COVID-19 vaccine's benefits continue to outweigh the risk of the uncommon side effect. <http://digit.in/jul21-51>



Tweaked yeast can make ethanol.

A genetically modified yeast can make ethanol from cornstalks and a harvest's other leftovers. <http://digit.in/jul21-52>

satellites in orbit and even more are planned to be launched in the coming years. As just one example, SpaceX has launched more than 1,700 of its Starlink satellites into space in the last two years, which has almost singlehandedly doubled the active number of satellites in orbit. And they have plans to launch thousands more in the next few years. They are not alone, there are plans for tens of thousands more to be launched by commercial companies, nation states or groups of nations into Low Earth Orbit (LEO satellites typically orbit between 500-1,000km above the Earth) and it is going to get very congested very soon.”

When it comes to introducing deorbital technologies and reducing risk of collisions in space, Mark tells us, “Inmarsat is committed to sustainability on Earth and in space, and already follows the rules around the safe management and decommissioning of our spacecraft. In addition to signing up to the Space Safety Coalition, we have been an Executive Director of the Space Data Association (SDA) since its inception in 2009, which works with other satellite operators. By sharing critical data on satellite positions and movements, and promoting best practice and responsible operations, the SDA works to reduce the probability of collisions and to make space opera-



Credit: Inmarsat

An artist's impression of Inmarsat's moon constellation

tions safer and more reliable. It is clear that the sustainability impact of the systems being deployed needs to be very carefully assessed and clear rules agreed for the operations of such systems, including a principle of 'leave no trace'. The parallels with climate change are clear, and as a space community we should learn the lessons that proactive and early management is required to ensure that we don't wait until the damage is done. We are entering a period of rapid deployment of space based services and capabilities, but it is incumbent on us all to make sure it is done in a way that ensures future generations can take benefit from access to space as well. Space is a precious resource and needs to be treated as such. A

'gold rush' approach of land grabs and over population of systems risks that critical sustainability. Careful management, clear regulations, robust debris mitigation capabilities and cooperation between operators and nations is vital.”

Inmarsat's capabilities are extending beyond the planet, and the company is exploring how to build up satellite internet infrastructure, on the Moon. Mark tells us, “providing communication and navigation capabilities around the Moon is a major challenge but one that will deliver benefits for us all as our species pushes out into space. Inmarsat will be a key member of a new consortium led by Telespazio to study the development of a satellite navigation and communications network that supports future missions to the Moon. It is funded by the European Space Agency (ESA) and the project will explore creating a single system that would serve all lunar missions, drastically reducing the cost of each trip that would no longer need its own infrastructure to be developed. It will also enable missions to land anywhere on the Moon and allow Moon rovers and other lunar installations to be operated from here on Earth. We could even see radio astronomers set up observatories on the far side of the Moon and to see further into the Universe than ever before.” 



Credit: Inmarsat

The Inmarsat GX5